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BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

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Department Of Computer Science & Engineering Data Science

Seminar Report on

“Pie Charts in Dynamics”

For the course:

DATA VISUALIZATION WITH POWER BI– 22CD633

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1. Introduction

Data visualization is an important method for presenting information in a clear and understandable way. Among the various visualization techniques, pie charts are widely used to show the proportion of different categories in a dataset. However, traditional pie charts display only a single moment in time and cannot show how data changes.

Dynamic pie charts are an advanced form of visualization that allows users to observe how proportions change over time or under different conditions. They use animation and interaction to present evolving data in a more engaging and meaningful way.

Dynamic pie charts help organizations and researchers understand trends, identify patterns, and make better decisions based on changing data.

2. What Are Dynamic Pie Charts

Dynamic pie charts are graphical representations that show changes in data proportions over time or across scenarios. Unlike static pie charts, dynamic pie charts evolve continuously to display variations in data.

They can be:

- Time-based — showing changes across months or years
- Interactive — allowing users to filter or compare data
- Animated — displaying smooth transitions between datasets

Dynamic pie charts help users visualize how data shifts and trends develop, making them useful for analysis and presentations.

3. Importance of Dynamic Visualization

Dynamic visualization is important because it helps users understand complex data quickly and accurately. It allows organizations to monitor changes and respond to new situations effectively.

Benefits of Dynamic Pie Charts

1. Capture Trends

They help identify patterns and trends in data over time.

2. Reveal Changes

They highlight sudden increases or decreases in values.

3. Support Decision Making

They provide visual insights that assist managers and analysts in making informed decisions.

Dynamic visualizations make data more meaningful and easier to interpret compared to static charts.

4. Choosing Data for Dynamic Pie Charts

Not all types of data are suitable for dynamic pie charts. The data should represent proportions and show meaningful changes over time.

Suitable Data Types:

1. Proportional Data

Data that represents parts of a whole.

Example:

Revenue by department, Budget allocation

2. Time-Series Data

Data collected over different time periods.

Example:

Monthly sales, Yearly population changes

3. Scenario-Based Data

Data comparing different situations.

Example:

Before and after analysis, Forecast vs actual results

Selecting the correct data ensures that the visualization is useful and meaningful.

5. Tools Used for Creating Dynamic Pie Charts

Several software tools are used to create dynamic pie charts. These tools provide features such as animation, interactivity, and data analysis.

Common Tools:

Power BI - A business intelligence tool used for creating interactive dashboards and visual reports.

Tableau - A data visualization tool that supports animated charts and storytelling.

Python (Plotly) - A programming tool used for creating advanced and customizable visualizations.

D3.js - A JavaScript library used to build highly interactive web-based charts.

6. Best Practices for Designing Dynamic Pie Charts

To create effective dynamic pie charts, certain design principles should be followed.

Design Guidelines

Limit the Number of Categories - Use only 5 to 7 slices to maintain clarity.

Use Consistent Colors - The same category should always have the same color.

Apply Smooth Animation - Transitions should be smooth to avoid confusion.

Highlight Important Changes - Important changes should be clearly marked.

7. Applications of Dynamic Pie Charts

Dynamic pie charts are used in many industries to analyze and present data.

Real-World Applications

Business - Used to track market share and sales performance.

Government - Used to show budget allocation and spending.

Sports - Used to analyze team composition and performance.

Healthcare - Used to monitor patient data and hospital resources.

8. Advantages of Dynamic Pie Charts

- Easy to understand
- Shows changes over time
- Interactive and engaging
- Improves decision-making
- Enhances data presentation

9. Limitations of Dynamic Pie Charts

- May become confusing with too many categories
- Requires software tools
- Needs proper design to avoid misinterpretation
- Animation may slow performance on low-end devices

10. Conclusion

Dynamic pie charts are powerful visualization tools that help present data in an interactive and meaningful way. Unlike traditional pie charts, they allow users to observe how data changes over time and make comparisons between different categories. These charts improve understanding of trends, support better decision-making, and enhance communication of information in business, education, healthcare, and many other fields.

Overall, dynamic pie charts make data visualization more engaging, accurate, and useful for analyzing real-world problems. By using appropriate data, proper design techniques, and suitable tools, dynamic pie charts can effectively transform complex data into clear and actionable insights.

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